

Periodicity

1. (P1) (O1) Let $P \in \mathbb{Z}[x]$ and $a_1 = 1$, with $a_{n+1} = P(a_n)$. Define $b_n \equiv a_n \pmod{N}$. Show that b_n is eventually periodic.
2. (P2) (O2) Let $a_1 = 1$ and $a_{n+1} = a_n^2 + a_n - 1$. Show that

$$\gcd(n, a_n) = 1$$

for all n .

3. (P3) (O1) Let $p(n)$ be the greatest prime divisor of n . Define the sequence by $a_1 \in \mathbb{N}$ and

$$a_{n+1} = p(a_n) + 1.$$

Show that a_n is eventually periodic and determine the length of the period.

4. (P4) (O2) Let

$$\text{rad}(n) = \prod_{p|n} p.$$

Show that the sequence defined by $a_1 \in \mathbb{N}$ and

$$a_{n+1} = \text{rad}(a_n) + 1$$

is eventually periodic and determine the length of the period.

5. (P5) (O3) Let $P \in \mathbb{Z}[x]$ and $a_1 \in \mathbb{Z}$, with

$$a_{n+1} = P(a_n).$$

Suppose that the sequence (a_n) is periodic. Show that the length of the period L satisfies $L \leq 2$.